

Article

Variation and Driving Mechanisms of Bark Thickness in *Larix gmelinii* under Surface Fire Regimes

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Abstract: Bark is vital for woody plants, providing protection, transporting nutrients and water, and storing essential resources. For fire-prone ecosystems, bark thickness is a key adaptive trait conferring fire resistance. Few studies have been conducted on the drivers of variation in bark thickness of the widely distributed *Larix gmelinii* (Rupr.) Kuzen in the Great Xing'an Mountains region, on the southern edge of East Siberia, where surface fire disturbances are frequent. To elucidate the relationships between variation in bark thickness (inner vs. outer bark) of *L. gmelinii* and plant size, environmental factors, and co-variation with other fire-tolerance traits, we selected 26 sites to set up plots and carried out a survey and bark sampling. Results showed that stem diameter primarily determines variation in bark thickness, especially outer bark. The proportion of outer bark to total bark increased accordingly as the tree increased in size. We also observed stronger correlated variation in outer bark thickness, tree height, and self-pruning capacity, implying that larger trees have thicker protective outer bark and taller heights with greater self-pruning, mitigating crown fire risks. Environmental factors appear to have a relatively limited effect on changes in bark thickness in *L. gmelinii*. Mean air temperature, annual precipitation, and total soil nitrogen content had some effect on outer bark thickness, and mean air temperature had some effect on inner bark thickness.

Keywords: bark thickness; bark ratio; fire adaptative strategy; surface fire regime



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1. Introduction

Bark plays a vital role in protecting the stems of plants, with a complex organization of living and dead cells outside the vascular formation layer [1]. Bark comprises two primary components, each with distinct origins, structures, and functions: the inner bark and the outer bark. The inner cortex primarily functions in the transportation, storage, and processing of photosynthesis and secondary compounds [2]. The outer bark serves several key functions, such as minimizing water loss, shielding against pathogen intrusion, providing mechanical stability, guarding against physical harm, insulating the trunk from severe weather conditions (such as extreme cold), and acting as a barrier against forest fires [3,4]. The thickness of bark determines the distance between external elements and vital tissues like the cambium [5]. The substantial investment required for thick bark probably propelled its evolution in ecosystems where even minor differences in thickness significantly improved fitness, leading to enhanced survival or increased reproductive success [5–7]. Therefore, understanding the intrinsic driving factors behind variations in bark thickness can assist us in exploring higher-order ecological issues, such as population fitness [7,8]. Moreover, since the inner bark and outer bark have different functional roles, clarifying the proportionality of their thickness is also a matter of urgent research.

Wildfires are among the most pervasive natural disturbances in forest ecosystems [9,10]. In boreal forests, the warming climate lengthens the fire season and has exacerbated fire activity, resulting in tree mortality and forest decline [11–13]. The boreal forest covers approximately 30% of global forest area, with over 70% located in Eurasian Siberia [14]. Low to moderate intensity, non-stand replacing (NSR) surface fires are predominant in Siberian boreal forests [15]. Surface fires burn loose needles, moss, and herbaceous vegetation that are at or near the surface of the ground, mostly by flaming combustion. The spread of surface fires is usually fueled by a layer of grass or deadfall. It has been found that pine trees exhibit a variety of characteristics associated with different fire regimes [16,17]. Pine species with a thick basal bark are predominantly found in surface-fire ecosystems [18]. They typically thrive in highly productive environments, tend to have considerable height, and possess flammable needles, leading to the self-pruning of dead branches. Consequently, these traits are correlated evolutionarily [17,18], defining what is known as fire-tolerant pine syndrome [17]. The Great Xing'an Mountains are part of the southern edge of the eastern Siberian boreal forest, with a fire regime also dominated by non-stand replacing fires. *Larix gmelinii* (Rupr.) Kuzen is the primary species in the Great Xing'an Mountains region [19], with taller and more self-pruning trees that possess tolerance to surface fires.

The growth of plant organs is influenced by geographical distribution, soil, climate, and other environmental factors [20,21]. Bark thickness fundamentally arises from growth and the accumulation of biomass. Apart from the impact of individual size on bark thickness, we hypothesize that diverse hydrothermal conditions and soil nutrient variations may induce variability in bark thickness. In fire-prone ecosystems, variation in bark thickness among species and communities can be explained by fire frequency. There is substantial evidence within extant species that demonstrates variations in bark thickness among populations and among closely related species inhabiting diverse fire regimes [18,22,23]. Elsewhere (e.g., in rainforest ecosystems), this investment is considered unnecessary, and large differences in bark thickness are more likely to be due to other environmental factors [24]. Research has found that no single environmental variable (such as climate or geography) can explain the differences in bark thickness [25]. Therefore, biophysical variables such as topographic features, soil nutrient and water availability, and climate need to be taken into account when assessing factors affecting bark thickness.

Through an extensive bark sampling and analysis initiative conducted in the Great Xing'an Mountains region, the three specific research objectives are as follows: (1) To investigate the correlation between total bark thickness, outer bark thickness, and inner bark thickness concerning plant size as well as examining how the ratio of inner/outer to total bark varies relative to plant size. (2) To examining the co-variation of bark thickness with other fire-tolerance traits. (3) To determine the most significant environmental factors contributing to thickness variation when the effect of plant size is considered. In the context of the boreal forest, where climate-induced fires have intensified, comprehending the drivers of variations in dominant species' bark thickness and the environmental factors that regulate major tree species in fire-prone regions becomes pivotal. This understanding will aid in exploring the role of key functional traits in influencing population fitness, tree resistance, and population maintenance.

2. Materials and Methods

2.1. Study Area and Sites Selection

The Great Xing'an Mountains region (50°11'–53°33' N, 121°12'–127°00' E) has a higher trend in elevation northward and lower southward, with its main mountain range oriented northeast–southwest [26]. The wide river valleys in the region form extensive areas of permafrost in meadows and marshes. Winters in the region can last up to 9 months with an average temperature below 10 °C, while summers typically last less than 1 month with an average temperature of approximately 22 °C. Over the past 40 years (1981–2020), the average annual temperature in the region was −4.3 °C and the average annual precipitation was 484.1 mm [27]. The soils in the region are predominantly dark-brown soil with relatively

rich surface organic matter and high humus accumulation [28]. During the 2021–2022 field survey, we selected 26 stand sites in the region. These stand sites were dominated by *L. gmelinii* and had experienced a single surface fire event between 1987 and 2001, characterized by low to moderate intensity. The years of the fires and the corresponding numbers of sites were: 1987 (9 sites), 1990 (2 sites), 1992 (3 sites), 1994 (3 sites), 1998 (2 sites), 1999 (2 sites), 2000 (2 sites), and 2001 (3 sites). Apart from *Larix gmelinii*, canopy species mainly consisted of *Betula platyphylla* (Sukaczew) and *Pinus sylvestris* L. var. *mongolica* Litv. The understory layer was primarily composed of *Ledum palustre* Linn, *Vaccinium vitis-idaea* L., *Spiraea salicifolia* L., *Rhododendron tomentosum* (Harmaja), etc. At higher elevations, *Pinus pumila* (Pall.) Regel tended to show greater coverage in the plots (Figure 1).

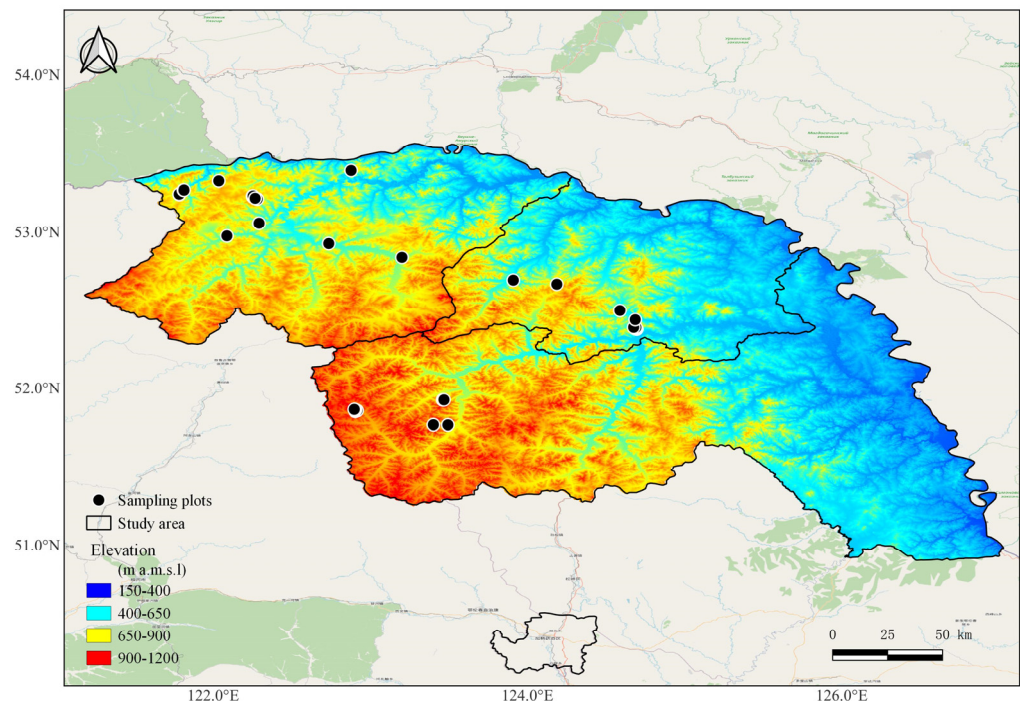


Figure 1. Distribution of sampling stand sites of *Larix gmelinii* tree bark.

2.2. Field Methods

At each stand site, sampling plots were randomly established, with sizes of 30 m × 30 m, based on stand size and habitat heterogeneity. The stand exhibits high habitat heterogeneity due to the local topography's diversity and the complex nature of its fire behavior. The horizontal distance between two plots was at least greater than 30 m. A total of 73 plots were established. We surveyed all trees (with a diameter at breast height ≥ 3 cm) in these plots, measuring their diameter at breast height (DBH), stem diameter (SD; measured at 0.3 m above the ground), tree height (TH; measured using the Vertex III hypsometer from Sweden), and self-pruning capacity (SP; height of lowest branch) on the trunk [29]. To ensure uniform sampling, each plot was divided into diameter classes at 5 cm intervals according to the DBH range, starting from a lower limit of 3 cm diameter. Approximately 30% of the trees in each diameter class were then randomly chosen for bark extraction. Bark samples taken had not been destructively damaged by fire except for blackening. A total of 274 sets of bark samples were collected. We used vernier calipers to measure the inner bark and outer bark thickness in the field. For *Larix gmelinii*, the total bark thickness equals the inner bark thickness plus the outer bark thickness [30] (Figure 2d).

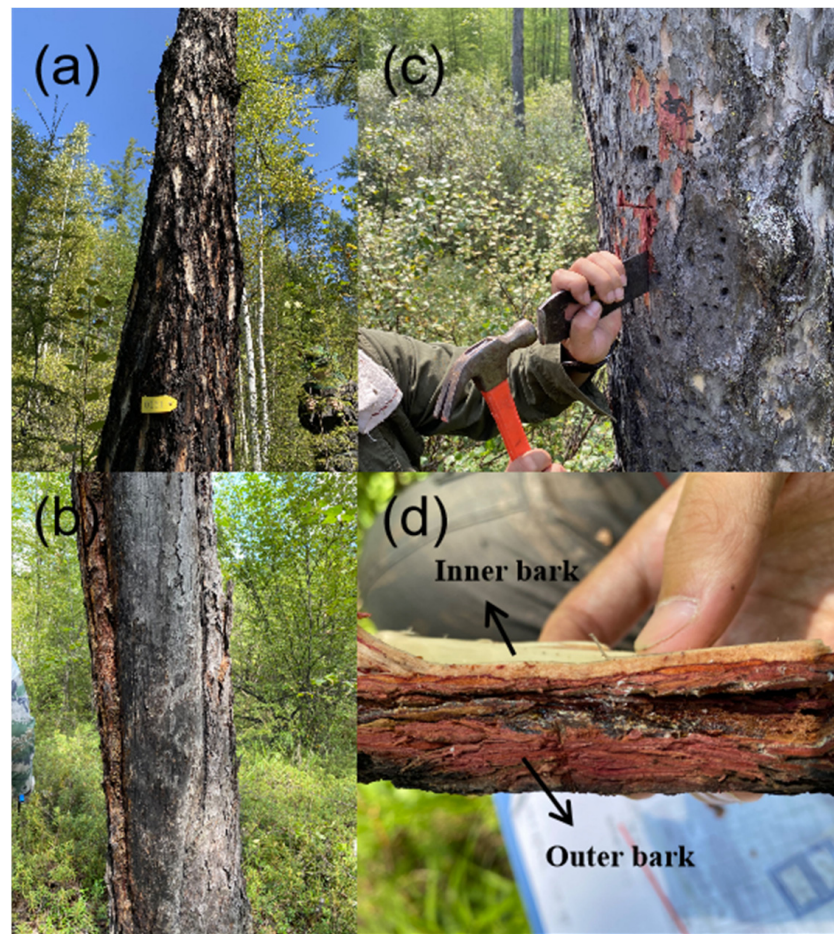


Figure 2. (a) A blackened *Larix gmelinii* trunk; (b) Bark suffering from fire damage to the trunk, but the tree survived; (c) Sampling at 30 cm from the ground; (d) The inner and outer barks of *Larix gmelinii* are distinctly colored in the figure: the beige portion of the bark is the inner bark and the reddish-brown portion of the bark is the outer bark.

2.3. Environmental Factor Surveys

When establishing the plots, we also measured and recorded slope and residual thickness. Slope was measured using a slope synthesizer (CN103063195A, by Institute of Mountain Hazards and Environment IMHE of CAS in Chengdu, China). Residual thickness was measured as depth to mineral soil, bedrock, or permafrost in small soil pits at three random locations per plot and then averaged [31]. Soil from three randomly chosen sampling points per sample plot was drilled and combined, then taken to the laboratory for further processing. During the sampling procedure, any weeds, dead branches, or surface leaves present on the soil were initially cleared. The soil was extracted from a depth of 0–10 cm below the ground surface as the experimental soil sample, with each sample weighing approximately 500 g. Kelvin's double-titration method was used for the determination of total nitrogen (TN) content, while the sodium hydroxide melting–molybdenum antimony anti-colorimetric method was used for the determination of total phosphorus (TP) [32]. Climatic factors including 1 km-resolution mean temperature (MT) and annual precipitation (AP) from 1970 to 2000 were extracted using Worldclim version 2.1 bioclimatic variables [33]. For the data analysis process, the environmental variables of each bark sample correspond to the environmental factors of the plot where the sample was located.

2.4. Data Analyses

We examined the association between bark thickness traits and plant size through univariate linear regression models, assessing the goodness of fit and significance of the models and their coefficients. To classify the samples based on stem diameter (SD), we employed a 10 cm diameter class (ranging from 5 cm as the lower limit). This classification divided all samples into six stem diameter classes (I: $5 \leq \text{SD} < 15$, II: $15 \leq \text{SD} < 25$, III: $25 \leq \text{SD} < 35$, IV: $35 \leq \text{SD} < 45$, V: $45 \leq \text{SD} < 55$, VI: $55 \leq \text{SD} < 65$; unit/cm). Within each diameter group, we averaged three bark trait indices (inner bark, outer bark, and total bark thickness) for the samples. Subsequently, we separately calculated the thickness of the inner bark and the proportion of the outer bark thickness to the total bark thickness.

Employing the Pearson correlation test, we identified correlations between bark traits, tree height, and self-pruning. The combined variations in tree height and self-pruning concerning bark traits might indicate individual tree resistance to fire. Furthermore, we investigated the co-variation of fire resistance traits through principal component analysis. We further clarified trends in the performance of bark thickness in relation to other fire-tolerant traits by fitting outer bark thickness to tree height and self-pruning using the logarithm function, and inner bark to tree height and self-pruning using the univariate linear fit.

Finally, we conducted an investigation into the influence of environmental predictors on bark thickness using a mixed-effects model. This model incorporated delineated diameter class as a random factor and included slope, residual thickness, mean temperature, annual precipitation, soil total nitrogen content, and soil total phosphorus content as fixed factors. Prior to this analysis, we compared the impact of the drivers of these two elements on bark thickness, using SITE and SD.CLASS as random effects, respectively, revealing a distinct and prominent influence of individual size as a driver.

3. Results

3.1. Associations between Bark Thickness Traits and Plant Size

Bark thickness traits showed a very close correlation with stem size (Figure 3a–c). Stem diameter (SD) explained 59% of the variation in total bark thickness (TBT) (Figure 3a), 32% in inner bark thickness (IBT) (Figure 3b), and 57% in outer bark thickness (OBT) (Figure 3c). OBT and TBT demonstrated faster growth with increasing SD (OBT~SDslope: 0.83; TBT~SDslope: 0.88), while IBT grew slower (IBT~SDslope: 0.05).

For total bark samples collected from the same adult tree at identical sampling heights, the proportion of total bark thickness attributed to the inner bark was notably smaller than that of the outer bark (Figure 2d). As the diameter class increased, the proportion of IBT decreased while the proportion of OBT increased.

3.2. Covariance of Bark Thickness with Other Fire Resistance Traits

Except for SD, bark thickness (both inner and outer bark) exhibited a positive correlation with both tree height and self-pruning (Figure 3a). The correlation coefficients of OBT with tree height were 0.61 ($p < 0.001$) and with self-pruning 0.53 ($p < 0.001$). The correlation coefficients of IBT with tree height were 0.5 ($p < 0.001$) and with self-pruning 0.32 ($p < 0.001$). The first and second principal components together explain 81.6% of the proportion of variance in the data, with the first principal component explaining 67.3% and the second 14.3% (Figure 4b). Additionally, it is evident that the angle of OBT versus tree height and self-pruning is larger compared with the angle of IBT versus tree height and self-pruning, further indicating a stronger correlation.

Figure 5 shows that the relationship between outer bark thickness and tree height and self-pruning branches is distributed as a logarithmic function with a good fit (TH~OBT $r^2 = 0.46$, $p < 0.001$; SP~OBT $r^2 = 0.38$, $p < 0.001$). The relationship between inner bark thickness and tree height and self-pruning branches was relatively dispersed, as seen from the distribution sample points, with a relatively poor linear fit for inner bark thickness and self-pruning (SP~IBT $r^2 = 0.09$, $p < 0.001$).

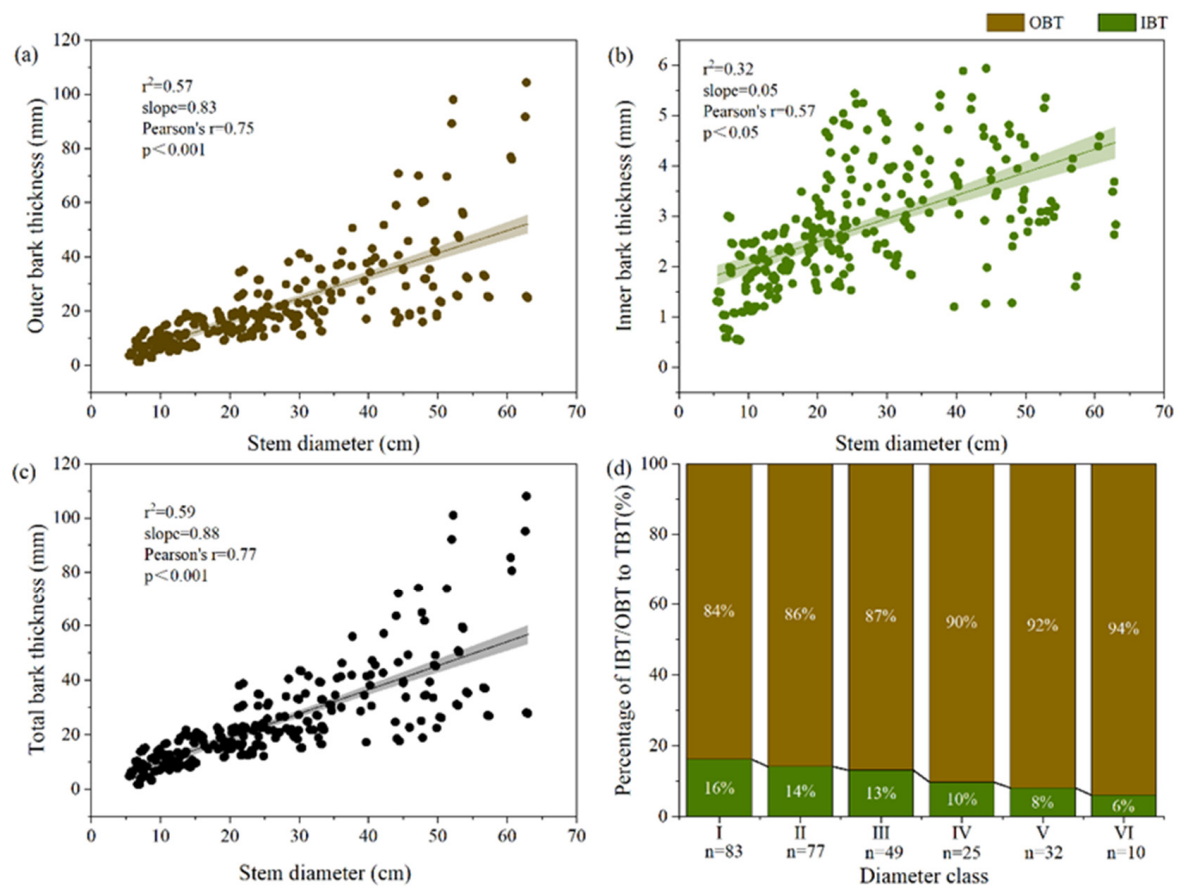


Figure 3. Bark thickness is strongly predicted by stem diameter. Regressions of (a) outer bark thickness, (b) inner bark thickness, and (c) total bark thickness against stem diameter are presented. Solid lines represent linear fits, while the shaded areas indicate the 95% confidence intervals. (d) IBT/OBT percentage of total bark thickness in each diameter class. Brown for outer bark, dark green for inner bark.

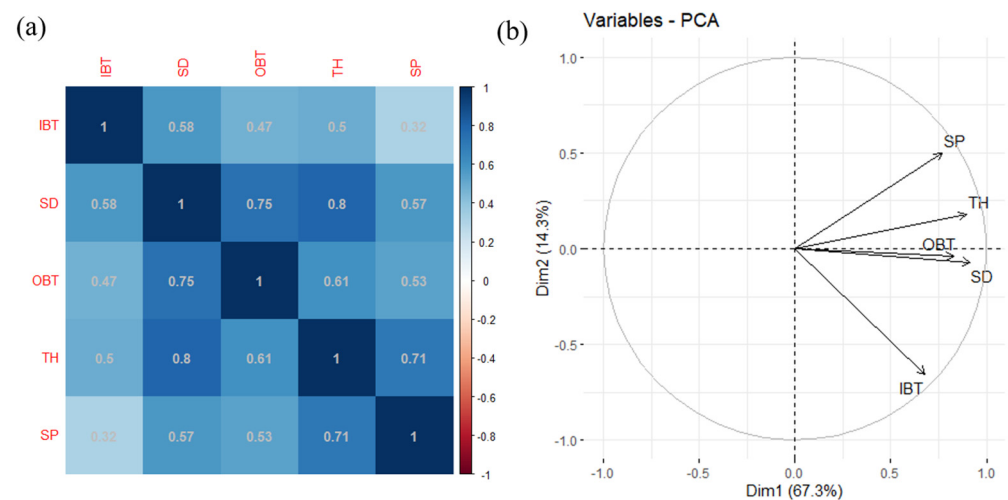


Figure 4. (a) Correlation tests for bark-related fire-resistance traits. (b) PCA demonstrates the relationship between variables and principal components, as well as the association between variables. SD indicates stem diameter, TH indicates tree height, SP indicates self-pruning, TBT indicates total bark thickness, IBT indicates inner bark thickness, OBT indicates outer bark thickness.

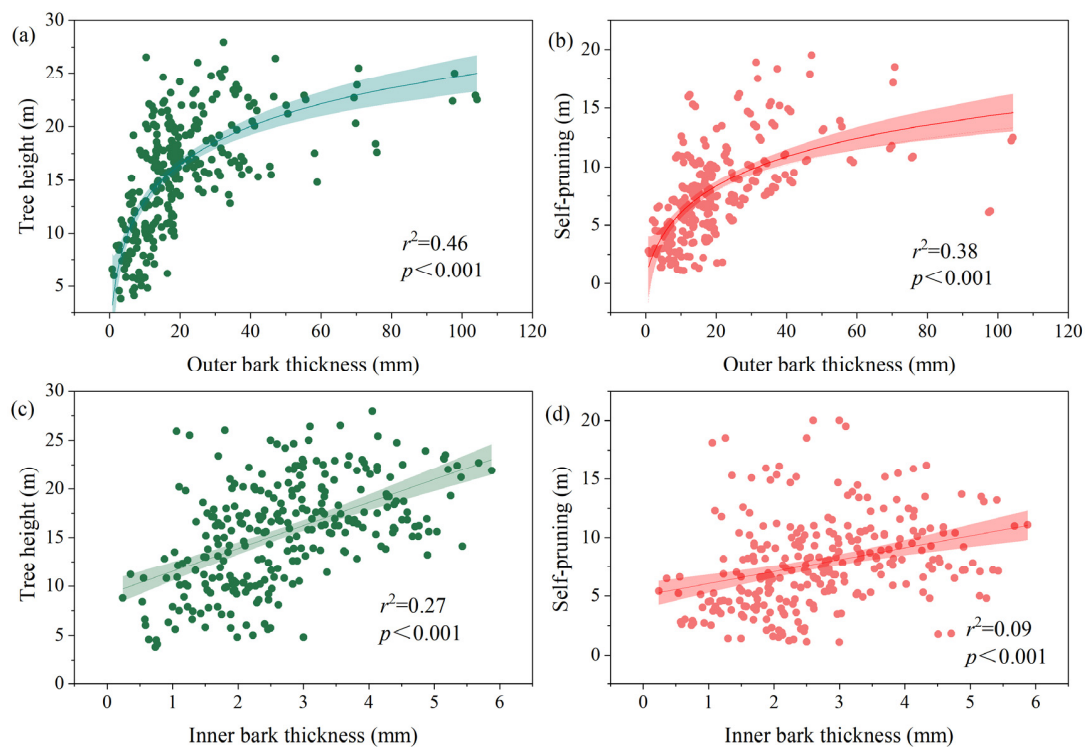


Figure 5. Outer bark thickness fitted with (a) tree height, (b) self-pruning logarithm function. Linear fit of inner bark to (c) tree height (d) self-pruning; confidence interval, 95%. Dark green for tree height, pink for self-pruning.

3.3. Roles of Environmental Factors in Explaining Bark Thickness Traits

We fitted models to predict inner and outer bark thickness using SITE and SD.CLASS as random factors, respectively (Table 1). When utilizing SD.CLASS as a random effect with OBT as the response variable, the relative contribution of the random effect to the overall variance was 65.3%. For IBT as the response variable, the relative contribution of the random effect to the overall variation was 42.8%. On the other hand, when SITE was used as the random effect with OBT as the response variable, the relative contribution of the random effects to the overall variance of OBT was 15.7%. Similarly, for IBT as the response variable, the relative contribution of the random effect to the overall variance was 5.5%. Using SD.CLASS as a random effect evidently offers more explanatory power compared with using SITE as a random effect.

Table 1. The variance and explanatory rate of bark thickness variability considering plot as a random effect and considering diameter class as a random effect, respectively.

Random Effects: OBT			Random Effects: IBT		
Random Effects	Variance	Standard Deviation	Random Effects	Variance	Standard Deviation
SD.CLASS (Intercept)	254.4	15.95	SD.CLASS (Intercept)	0.6002	0.7747
Residual	135.3	11.63	Residual	0.8192	0.9051
Relative contribution of the random effect: 65.3%			Relative contribution of the random effect: 42.8%		
SITE (Intercept)	45.1	6.716	SITE (Intercept)	0.08332	0.2887
Residual	241.8	15.549	Residual	1.36059	1.1664
Relative contribution of the random effect: 15.7%			Relative contribution of the random effect: 5.5%		

Slope, residual thickness, mean temperature, annual precipitation, soil TP content, and soil TN content were included as fixed effects, and SD.CLASS was included as a random effect in the model, as shown in Table 2. The fixed effects that significantly influenced

the response variable OBT include mean temperature (estimates = 0.27; $p < 0.001$), annual precipitation (estimates = 0.06; $p < 0.016$), and soil total nitrogen content (estimates = 4.9; $p = 0.043$). This model exhibited a marginal R^2 of 0.044 and conditional R^2 of 0.723. Conversely, only mean temperature (estimates = 0.05; $p < 0.001$) significantly affected the response variable IBT, with a marginal R^2 of 0.084 and conditional R^2 of 0.498.

Table 2. Mixed-effects model predicting inner and outer bark thickness based on site conditions, soil nutrients, climatic factors.

Random Effects: OBT				Random Effects: IBT			
Predictors	Estimates	CI	p	Predictors	Estimates	CI	p
(Intercept)	83.51	44.8–122.21	<0.001	(Intercept)	3.01	0.14–5.88	0.04
slope	−0.38	−0.67–0.08	0.734	slope	0.01	−0.02–0.03	0.52
Residual thickness	−0.39	−0.84–0.07	0.413	Residual thickness	0.03	−0.01–0.07	0.098
Mean temperature	0.27	0.14–0.4	<0.001	Mean temperature	0.05	0.03–0.08	<0.001
Annual precipitation	0.06	0.01–0.11	0.016	Annual precipitation	0.00	−0.01–0.01	0.201
Soil TP content	0.16	−11.16–11.48	0.191	Soil P content	0.47	−0.82–1.75	0.185
Soil TN content	4.9	−1.6–11.53	0.043	Soil N content	0.47	−0.82–1.75	0.071
$N_{SDCLASS:6}$		Conditional $R^2 = 0.723$		$N_{SDCLASS:6}$		Conditional $R^2 = 0.498$	
Marginal $R^2 = 0.044$				Marginal $R^2 = 0.084$			

Note: “ $N_{SDCLASS}$ ” was the number of random effects groups.

4. Discussion

4.1. *Larix Gmelinii* Bark Thickness Was Mainly Driven by Plant Size

Our bark sampling of *Larix gmelinii* revealed a significant variation in both total bark thickness (TBT) and outer bark thickness (OBT), primarily associated with stem diameter (SD), aligning with similar observations in other tree species [7,20,34]. SD has been demonstrated to globally explain 72% of the variation in TBT, displaying a stronger explanatory rate compared with our study’s findings [1]. Plant size is one of the most important indicators of tree growth and development; OBT and IBT of *L. gmelinii* were significantly correlated with SD (Figure 3). In our study, we found that for *L. gmelinii*, OBT exhibited a stronger correlation with TBT than inner bark thickness. Furthermore, OBT constituted a larger proportion of TBT. This proportion increased with individual size. The relatively higher carbon investment in the outer bark of *L. gmelinii*, particularly evident during the plant’s growth and development, seems closely linked to the fire-prone environment of the region [5]. SD also provides a higher rate of explanation for outer bark thickness compared with inner bark thickness. The outer portion of bark is comprised of an accumulation of dead cells known as phellem (derived from the cork cambium) or rhytidome [4,35]. *L. gmelinii* bark has been proven to be a raw material for thermal insulation materials [1,36], but for *L. gmelinii* this depends mainly on the outer bark [37].

The developmental size of individuals and environmental factors are considered the primary influences on intraspecific variation [38]. We assessed the explanatory power of different developmental stages and environmental factors on inner and outer bark thicknesses, incorporating random effects for diameter class and sample site, respectively. One is considered an endogenous factor, while the other is exogenous. The results revealed that individual size, as an endogenous factor, exhibited stronger explanatory power for bark thickness.

4.2. Bark, Tree Height, and Self-Pruning Constitute Defenses against Surface Fire Traits Syndromes

Surface fire damage predominantly targets the phloem and the formation layer within tree trunks, with the extent of injury reliant on factors like fire intensity, duration, and thermal conductivity. Typically, the lower part of the trunk, closer to the ground, experiences more severe damage [39]. This frequent occurrence of low-intensity fire regimes serves as a prime example of selective pressures that favor the development of *L. gmelinii*’s thicker bark [40]. Simultaneously, our results demonstrate a covariate relationship between bark

and tree height, self-pruning, and especially outer bark, a trait syndrome that reflects *L. gmelinii*'s adaptive strategy to surface fire. For ecosystems where surface fires are the dominant fire regime, it is difficult for seedlings or saplings to survive. Nevertheless, as the plant matures and grows, the individuals become larger, leading to an increase in the proportion of outer bark thickness to total bark thickness. This suggests that thick bark is effective in protecting trees from surface fires [41], while the self-pruning capability of mature *L. gmelinii* trees, along with their towering height that occupies the canopy, minimizes the chance of fire penetration into the canopy. These fire-adaptive traits combine to make this tree highly resistant to surface fires. Prominent examples also are *Araucaria araucana* and *Fitzroya cupressoides* in the Andes, and *Sequoiadendron giganteum* and *Calocedrus decurrens* in western North America [42–44]. In certain high-intensity wood fire ecosystems, like shrublands where fire intensity is considerably intense, minor differences in bark thickness may not significantly contribute to fire protection. Consequently, thick bark might not be a selected trait in these environments, and the plant height tends to be lower. Examples of such ecosystems include Mediterranean shrubland ecosystems [45].

4.3. Environmental Factors Affecting Bark Thickness

Plant growth is influenced by environmental factors such as soil nutrients, temperature, and humidity. Previous studies have focused on height and radial growth of plants, and the environmental predictor variables of these studies have a high explanatory rate in that warmer, wetter, and more nutrient-rich environments significantly promote plant growth [46,47]. However, in our study, environmental factors exhibited lower explanatory power. Our modeling results revealed that mean temperature, annual precipitation, and soil total nitrogen content had relatively minor effects on outer bark thickness, despite their statistical significance. Similarly, mean temperature showed a significant but negligible impact on inner bark thickness (estimates = 0.05). These findings are similar to those of Rosell's global-scale study [1], which identified stem size as the primary driver of bark thickness variation, with environmental factors playing a lesser role. Moreover, investigations into bark thickness in major Chinese woody plants indicated a correlation between biomass accumulation and soil fertility, temperature, and seasonal precipitation fluctuations [21]. Indeed, species-level studies have mostly concluded that the genetic and developmental characteristics of plants can have a major impact on trait differentiation, sometimes outweighing the influence of environmental factors [20,48]. Even in ecosystems where fires are infrequent, some variability in bark thickness persists, probably influenced by other factors. For instance, in temperate rainforest ecosystems less susceptible to fires, bark investment primarily responds to factors such as soil resources, minimum temperatures, and seasonal water scarcity [24]. Additionally, for other plant organs, environmental factors significantly promote trait-driven variation. For instance, temperature and light contribute to variations in leaf area at both intraspecific and interspecific levels [49,50].

While past large-scale studies have indicated the significant influence of fire regimes on bark thickness [5], our study did not incorporate fire as a predictor of bark thickness in *L. gmelinii* due to the absence of site-scale data on mean fire return interval. In future studies, we aim to integrate fire intervals into our model through methods such as reconstructing fire histories. Additionally, as a dominant species in the boreal forests of East Siberia, further comprehensive exploration is required to understand the physiological functions of *L. gmelinii* bark under cold and fire-prone conditions.

5. Conclusions

Following the comprehensive sampling and extensive analysis of *L. gmelinii* bark across the expansive Great Xing'an Mountains region, our investigation revealed that stem diameter predominantly governs the variability in bark thickness, particularly in the outer bark thickness. Notably, as the size of the trees increased, the outer bark's proportion within the total bark exhibited a proportional augmentation. Concurrently, we observed a stronger correlated variance in the outer bark thickness, aligning directionally with tree

height and self-pruning capability. This association implies that larger specimens not only develop thicker outer bark as a fire-resistant mechanism but also tend to exhibit greater tree height and enhanced self-pruning abilities, serving to mitigate the risk of crown fires. The influence of environmental factors on the variation in bark thickness of *L. gmelinii* appears to be relatively limited. Mean temperature, annual precipitation, and soil total nitrogen content exerted certain effects on the thickness of the outer bark, and mean temperature displayed a discernible impact on the inner bark thickness.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to the data involving a student's thesis.

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Conflicts of Interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

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